

Thesis Assignment

In an electric power system, the balance between consumption and generation must be maintained at all times. If this balance is not met, the system frequency will change; the initial rate of change of frequency is determined by the inertia present in the system. Inertia is inherently present in conventional synchronous generators with rotating masses, whereas units connected to the grid via power electronics practically provide no inertial contribution to frequency changes. Therefore, it is important to understand how the frequency gradient changes in the interconnected European synchronous system as inertia decreases.

The student's tasks are:

- To study the available literature on the effect of decreasing inertia on the rate of change of system frequency;
- To investigate, in a simulation environment, using a simplified network model concentrated on a single node representing both consumption and generation, the impact of reduced inertia on system frequency in the event of a system blackout.



Investigations on the rate of change of frequency

with a single machine-single busbar model

Bachelor's Thesis

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STUDENT DECLARATION

I, *András István Babai*, the undersigned, hereby declare that the present Bachelor's Thesis work has been prepared by myself and without any unauthorized help or assistance. Only the specified sources (references, tools, etc.) were used. All parts taken from other sources word by word, or after rephrasing but with identical meaning, were unambiguously identified with explicit reference to the sources utilized. In the Appendix, I have clearly indicated whether I used artificial intelligence tools in preparing the thesis; if so, I have specified the manner and extent of their use in the table. I acknowledge that I bear full responsibility for any content generated by artificial intelligence – regardless of the extent of its use.

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Budapest, December 12, 2025

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Abstract

Electric power generation has undergone significant changes in recent decades. Traditional synchronous generator-based production is increasingly being replaced by Renewable Energy Source (RES) such as photovoltaic (PV) power plants and wind turbine generators (WTGs). Although these technologies offer clear environmental benefits, their integration poses considerable challenges for transmission system operation. Their inherent unpredictability and inverter-based energy conversion lead to reduced system stability and reliability. These issues become even more critical during large disturbances or system separations, which may ultimately result in widespread blackouts. To address these challenges, it is essential to understand the behavior of power systems with high RES penetration under various operating scenarios. As part of a broader research project, the long-term goal is to develop a model capable of simulating the behavior of the Central European Synchronous Area during system splits.

This thesis serves as preparatory work for that objective and focuses on the modeling approaches of the two main RES technologies: photovoltaic power plants and wind turbine generators. It reviews the most important RES types, discusses their modeling possibilities, and provides a detailed description of the selected models. These models are then implemented in a single-node test system, designed to replicate the system separation event of 8 January 2021. The simulated frequency responses of various grid configurations are compared against measured data from the actual disturbance.

The case studies demonstrate that the role of RES extends far beyond being a destabilizing factor in the grid. With appropriate control strategies, RES units can significantly contribute to frequency support and overall grid stability.

Acronyms

aFRR	automatic Frequency Restoration Reserve. 8, 9
AVR	Automatic Voltage Regulator. 15, 37
CE	Central European. 4, 5
DFIG	Doubly Fed Induction Generator. 20
ENTSO-E	European Network of Transmission System Operators for Electricity. 5, 6, 34
FCR	Frequency Control Reserve. 8–10, 37
FFR	Fast Frequency Response. 8, 17, 18, 23, 26, 27, 29, 31, 33, 35
GOV	Turbine Governor. 16, 37
IBFFR	Inertial-Base Fast Frequency Response. 23–27
ICS	Incident Classification Scale. 3
mFRR	manual Frequency Restoration Reserve. 8, 9
PF	PowerFactory. 14, 17, 22
PMSG	Permanent Magnet Synchronous Generator. 21
PSS	Power System Stabiliser. 15, 37
REEC	Renewable Energy Electrical Control. 17, 23
REGC	Renewable Energy Generator/Converter. 17, 23
REPC	Renewable Energy Plant Control. 17, 23
RES	Renewable Energy Source. i, 13
RMS	Root Mean Square. 14
RoCoF	Rate of Change of Frequency. 6
SCIG	Squirrel Cage Induction Generator. 19
SEXS	Simplified Excitation System. 15
TSO	Transmission System Operator. 5, 10
WECC	Western Electricity Coordinating Council. 17–19, 22, 25, 29, 37
WRIG	Wound Rotor Induction Generator. 20, 21
WRSG	Wound Rotor Synchronous Generator. 21
WTG	Wind Turbine Generator. 8, 13, 14, 22–28, 32–35, 37

1 Preamble

1.1 Aims of the research

The current thesis forms part of a larger research project, whose final goal is to develop a detailed yet simple model that represents the generation, load, and cross-border flow capacity for each country and border within the Continental European Synchronous Area. The aim of this model is to serve as a practical tool capable of simulating any system separation at any physical border.

In the final model, each country will be represented by a single-node model. The aim of this thesis is to develop and test a single-node model that can later serve as an appropriate representation of any country. Traditionally, a single-node model consists of both generation and load components; however, the current thesis focuses primarily on the modeling of generation units (Section 3). The single node model presented in this thesis bases on a well known system separation, occurred on the 8th of January 2021.

The cross-border connections will be modeled using the resultant of all direct connections between the represented countries, but this topic will be addressed in a future paper.

1.2 Motivation

Our motivation for modeling and investigating system separations is divided into two parts.

1.2.1 Changing electric grid

Traditionally, the electric grid has been a centralized system, consisting of a few high-performance generation units and several predictable loads. The operational advantages of such a system are undeniable: it is robust, resilient, and resistant to most disturbances. Some of the main reasons behind these properties are the following:

- The output power of generation units is independent of the *weather* and the *time of day*.
- Load demands are predictable, depending primarily on the time of day.
- Power flow is unidirectional.

These advantages play a crucial role in the frequency stability of the grid and its resilience against possible system separations or large imbalances.

However, in the last few decades, several significant changes have occurred, fundamentally transforming the key properties of the electrical grid. Numerous photovoltaic plants and wind turbine generators have been connected to the grid. Today, the contribution of these units to energy generation is no longer negligible. For example, during the summer 2025, their total generation has exceeded the power output of synchronous machines several times. In these cases the significant ratio of inverter based generators weakens the advantages of the traditional synchronous generators. The key changes arising by this transition are the followings:

- Part of the producers have become **dependent** on *weather* and *time of day*, a proper generation planning requires accurate weather forecast

- Forecasting consumer demand has become more difficult, requiring **more accurate** weather predictions
- Power flow has become **bidirectional**

These tendencies are expected to further strengthen in the coming decades, which means that investigating their impact on frequency and grid stability is crucial for stable system operation. Current research aims to provide reliable answers to these concerns in terms of frequency stability.

1.2.2 Increasing system failures and malfunctions

Major disturbances, especially system blackouts, are considered rare events in the European grid. However, in recent years, numerous disturbances have occurred compared to what was previously usual. Table 1 provides a summary of the most significant disturbances that have occurred in recent years. The severity of each grid disturbance is rated according to the ICS [1].

Table 1 | Major European disturbances (2020-2025) [1]

Date	Affected countries	ICS scale	Consequence
2025.04.28	ES, PT	Scale 3	System blackout, total outage
2024.06.21	ME, BA, AL	Scale 3	System blackout, partial outage
2021.07.24	ES, PT	Scale 2	System blackout
2021.01.08	Southern and Central Europe	Scale 2	System blackout

The impacts of these blackouts and disturbances, on the one hand, cause significant inconvenience for the affected citizens, while on the other hand, they result in severe, often multi-billion euro damages to the economies of the affected areas.

2 Literature review

2.1 The European electric grid

In Europe, a large interconnected electricity system has been developed to ensure operational and supply security (Figure 1). In this system, the countries marked in blue are synchronized with each other (Central European (CE) synchronous area), and the countries marked in red are connected to them via a DC cable network. Inside the CE synchronous area, each country is connected to its neighboring countries via direct AC lines. Thus forming a large synchronized grid in which every local disturbance can have widespread effects. Therefore, improper handling of disturbances may lead to severe consequences, such as system separation or blackouts.

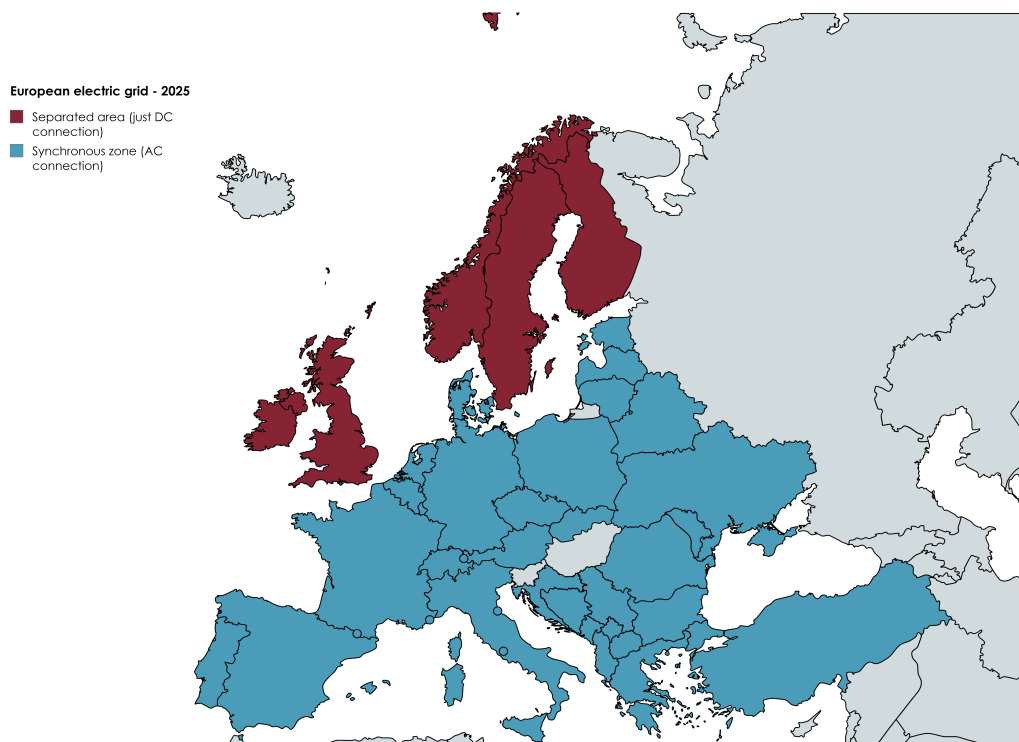


Figure 1 | European electric grid

2.2 Properties of electric grids

2.2.1 Power

The interconnected electric system establishes the possibility to transfer electrical energy between different countries. It means that the electricity produced in one country can be consumed in another or even on the other side of the continent.

On one hand, the interconnected system provides significant flexibility to every country within the grid. It is not necessary to rely solely on domestic generation, as countries have the opportunity to export and import electricity. This is particularly beneficial for renewable energy integration, where local imbalances between generation and consumption can be mitigated at the European level.

On the other hand, the interconnected system also carries several risks. For example, when equipment or transmission lines malfunction, some connections (paths of power flow)

may fail, while others can become overloaded. Such situations can easily lead to system separation, frequency collapse, or even blackouts.

2.2.2 Frequency

The *frequency* (f) is one of the most important characteristic of the AC electric grids. In the European interconnected system the nominal frequency value is $f = 50$ Hz. Keeping the frequency around its nominal value is fundamental, both from the generation and consumption points of view. In favor of maintaining the nominal frequency, all local Transmission System Operator (TSO) and the ENTSO-E specified strict regulations in term of allowable frequency ranges (Table 2).

Table 2 | Frequency ranges for the CE synchronous area [2]

CE	
FqA	± 50 mHz
FqC	200 mHz
FqE	800 mHz
FqF	49.00 Hz

Where **FqA** is the standard frequency range, this range is the symmetrical interval around the nominal frequency within which the system frequency of a synchronous area is supposed to be kept. **FqC** is the maximum steady-state frequency deviation - FCR full activation range; it is the target for maximum steady state frequency deviation after the dimensioning incident. **FqE** is the maximum instantaneous frequency deviation - minimum frequency; it is defined by the maximum instantaneous frequency deviation after the dimensioning incident. **FqF** is the demand disconnection starting mandatory level: - load shedding frequency level; it is specified in the Emergency and Restoration code (E&R).

The principal requirement for frequency stability is the continuous equilibrium between generation and load. If generation and consumption lose this balance, the system frequency begins to deviate. In the case of over-consumption, the frequency decreases, while in the case of over-generation, it increases (2.1).

$$\begin{aligned} \Delta P = P_m - P_e > 0 &\leftrightarrow f \uparrow \\ \Delta P = P_m - P_e < 0 &\leftrightarrow f \downarrow \end{aligned} \quad (2.1)$$

$P_m(t)$ — mechanical power (generation)

$P_e(t)$ — electrical power (consumption)

In order to maintain the balance between generation and load, constant power and frequency regulation are required. In practice it is realized via a multi-step frequency control scheme, which is presented in Section 2.5.

2.2.3 Power-frequency characteristic

The power-frequency characteristic is a crucial attribute of an electric grid. It basically describes the expected frequency change after a given amount of generation or load loss.

It is marked with the Greek letter *lambda* (λ), its dimension is $\frac{\text{MW}}{\text{Hz}}$. Its simplest definition is (2.2), where ΔP is the power imbalance that occurred during the imbalance, and Δf is the frequency deviation [3].

$$\lambda = \frac{\Delta P}{\Delta f} \quad (2.2)$$

Even though the definition of power-frequency characteristic seems to be simple, the way to calculate it is a bit trickier. The exact calculation method is reported in detail in [3].

The power-frequency characteristic is a global property of the whole interconnected system, however it can be determined for every region or country. Every country has a pre-determined *contribution coefficient*, marked with c_i . By multiplying the power-frequency characteristic of the overall system by these coefficients, the power-frequency characteristic of the given region can be calculated (2.3) [3].

$$\lambda_i = \lambda \cdot c_i \quad (2.3)$$

The values of c_i are determined by ENTSO-E on a yearly basis; the actual values dating back to 2021 are reported in [3].

2.2.4 Inertia

Inertia plays an extremely important role in the frequency stability of electrical power systems. The easiest approach to understand it is to consider it as a *moment of inertia* that slows down the speed change of rotating machines, thereby slowing down the rate of frequency change. In other words, it reduces the Rate of Change of Frequency (RoCoF) value. This moment of inertia traditionally comes from the rotating masses of the synchronous generators.

2.3 Role of inertia

As mentioned earlier, inertia is one of the key factors of electric grids in terms of frequency stability; therefore, a thorough understanding of it is essential. For this purpose, two definitions of inertia are presented in this section.

2.4 Inertia as rotational inertia

One way to define the term inertia in electric grids is as follows.

Newton - II.

The definition of inertia can be derived from Newton's second law:

$$F = m \cdot a = m \cdot \frac{dv}{dt} \quad (2.4)$$

Moving from the force dimension to torques, the (2.4) formulation is directly interpretable for electric grids.

$$\Delta T = T_m(t) - T_e(t) - D_p \cdot \omega = J \cdot \frac{d\omega}{dt} \quad (2.5)$$

where,

$$\begin{aligned}
 T_m(t) &— \text{mechanical torque [N}\cdot\text{m]} \\
 T_e(t) &— \text{electrical torque [N}\cdot\text{m]} \\
 D_p &— \text{damping factor from mechanical friction [N}\cdot\text{m}\cdot\text{s]} \\
 J &— \text{rotational inertia [kg}\cdot\text{m}^2] \\
 \omega = 2 \cdot \pi \cdot f &— \text{angular speed [rad/s]} \\
 t &— \text{time [s]}
 \end{aligned}$$

If the value of ΔT is not equal to zero, the angular velocity of the rotating masses changes, resulting in a variation of the electrical frequency of the grid. The rotational inertia (J) is directly expressible from (2.5), if the effect of damping factor is omitted (2.6) [4].

$$J = \frac{T_m(t) - T_e(t)}{\frac{d\omega}{dt}} \quad [\text{kg} \cdot \text{m}^2] \quad (2.6)$$

Kinetic energy

Another way to define inertia is to derive it from the kinetic energy (E_{kin}) of the rotational masses in the electric grid:

$$E_{kin} = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2} \cdot J \cdot \omega^2 \quad (2.7)$$

By rewriting the equation (2.7) in terms of power, the equation (2.8) obtained.

$$P_m(t) - P_e(t) = \frac{dE_{kin}}{dt} = \omega \cdot J \cdot \frac{d\omega}{dt} \approx \omega_n \cdot J \cdot \frac{d\omega}{dt} \quad (2.8)$$

$$\begin{aligned}
 P_m(t) &— \text{mechanical power of generators [W]} \\
 P_e(t) &— \text{electrical power consumed by loads [W]} \\
 \omega_n &— \text{nominal angular velocity [rad/s]}
 \end{aligned}$$

Combining (2.6) and (2.8), the following formulation is derived [4]:

$$J = \frac{T_m(t) - T_e(t)}{\frac{d\omega}{dt}} = \frac{P_m(t) - P_e(t)}{\frac{d\omega}{dt} \cdot \omega_n} \quad [\text{kg} \cdot \text{m}^2] \quad (2.9)$$

2.4.1 System equivalent inertia

The second way to interpret the meaning of inertia is through the definition of the so-called *system equivalent inertia*. This is defined as the ratio of the *total kinetic energy* E_{kin} to the rated power S of the given synchronous generator. Furthermore, by substituting equation (2.7), the relationship between the two definitions of inertia can be obtained as shown in (2.10) [4].

$$H = \frac{E_{kin}}{S} = \frac{J \cdot \omega_n^2}{2 \cdot S} \quad [\text{s}] \quad (2.10)$$

H – system equivalent inertia [s]

S – rated power of a synchronous generators [VA]

The first part of equation (2.10) is used in Section 3 to determine the inertia of the synchronous machine applied in the single node model. Meanwhile, equation (2.6) is used in Section 5 for the design of the Fast Frequency Response (FFR) in PV plants.

To summarize, the two most important representations of inertia, namely the *rotational inertia* J and the *system equivalent inertia* H , have been defined. Although both of them are used in the following sections, the consistency of the notation is ensured in order to avoid any misunderstandings.

2.4.2 Inertia of the electric grid

The previous definitions of inertia are defined for given synchronous machines. However, for the purpose of determining the inertia of an electric grid with several synchronous generators, the following formula should be used.

$$H_{\text{syn}} = \frac{\sum_i^N H_i \cdot S_{G,i}}{\sum_i^N S_{G,i}} \quad (2.11)$$

H_{syn} sometimes also called as *Inertia Constant*.

If the grid contains inverter based generation (PV plants, WTGs) as well, the formula ought to be modified according to (2.12) [4].

$$H_{\text{sys, gen}} = \frac{\sum_i^N H_i \cdot S_{G,i}}{P_{\text{load}}} \quad (2.12)$$

2.5 Frequency control

As we have seen in Section 2.2, the constantly varying power of loads results in the value of frequency being under constant variation as well. In order to maintain it around its nominal value (f_n), constant frequency control is required.

The ideal frequency control system is capable of handling frequency deviations caused by grid disturbances, in addition to those originating from natural load fluctuations. In practice, the short term frequency control (called *balancing service*) is implemented as a multi-stage process consisting of four distinct steps: FCR, aFRR, mFRR.

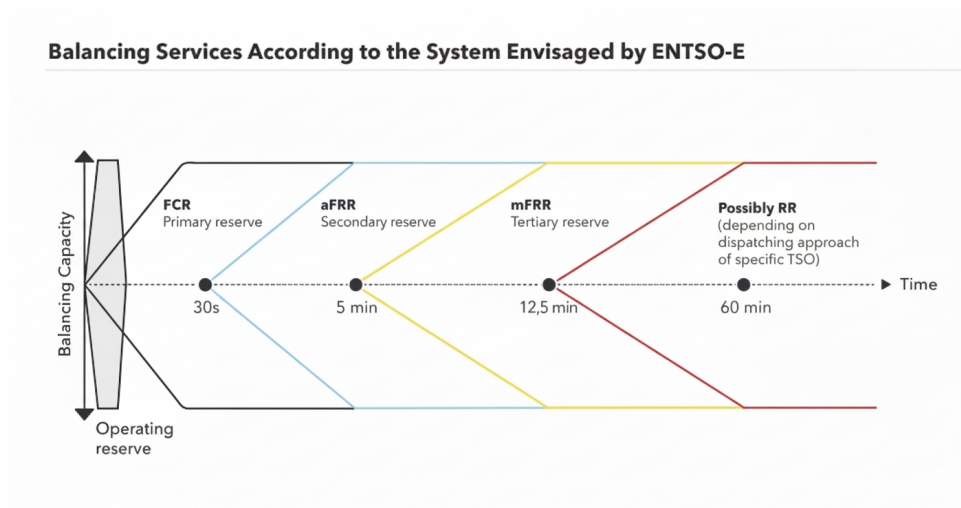


Figure 2 | Multi-step frequency control system [5]

Below a brief introduction is presented about each step of the balancing service, in order to understand the frequency regulation in the model presented in Section 3.

Frequency Control Reserve (FCR)

- *Alternative names:* primary control reserve
- *Activation time:* right after the disturbance, but no more than 30 seconds
- *Operation length:* should be capable to operate during at least 15 minutes
- *Activation method:* automatic, based on frequency measurements at the connection point of the FCR supplier
- *Trigger:* $\Delta f = \pm 0.01$ Hz
- *Goal:* slow down or if possible stop the change of frequency

automatic Frequency Restoration Reserve (aFRR)

- *Alternative names:* secondary reserve
- *Activation time:* at least 30 seconds after disturbance, but no more than 5 minutes
- *Operation length:* with no limit
- *Activation method:* automatic
- *Trigger:* after 30 seconds of FCR activation, the frequency still varies
- *Goal:* stop the frequency change, restore it to nominal value

manual Frequency Restoration Reserve (mFRR)

- *Alternative names:* tertiary reserve
- *Activation time:* at least 12.5 minutes after disturbance
- *Operation length:* at least 5 minutes

- *Activation method*: manual, by TSO
- *Trigger*: varying among TSO-s
- *Goal*: stop the frequency change, restore it to nominal value

Current thesis focuses only on the FCR control performed and controlled directly by the modeled power plants.

3 Single node model

The role of single node models in the field of electric grid simulations, especially in the area of **frequency stability studies**, is undeniable. Their most significant advantage is the capability of representing a power system of any complexity within a single node, thereby significantly reducing the number of variables and parameters that must be considered.

As it was mentioned in Section 1, in our future research plan, each country's generation and load will be represented by one single node model. Then the model of the whole European network will build up from these single node models by adding all the necessary connections.

3.1 The simulated event

In this paper, the European network was simplified by merging all the countries into two different single node models. The base of these models was the system split occurred on the 8th of January 2021 [6].

That day as a consequence of outages of several transmission network elements the synchronous area of Continental Europe was separated into two parts: *EAST*, *WEST*, see Figure 3.

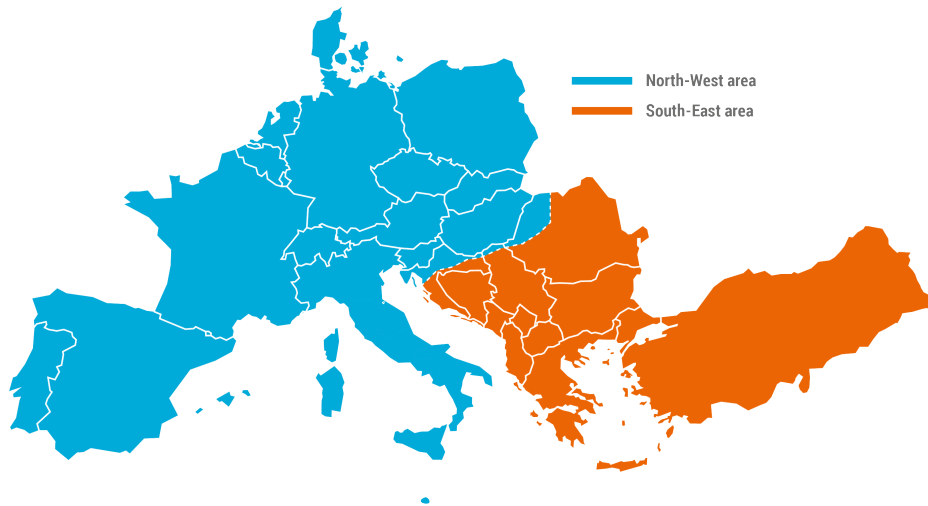


Figure 3 | Separation of Continental Europe Synchronous Area on 8 January 2021 [6]

At the moment of system separation, there was a significant power flow (6.3 GW) from EAST to WEST. Consequently, immediately after the separation, this amount of generation was missing on the WEST side, while the EAST side experienced an over-generation of 6.3 GW. These imbalances have a severe effect on grid frequency stability, therefore their proper investigation is justified.

3.2 Model structure

3.2.1 The initial model

Our model is based on a simple single node model, consisting only synchronous generators. A detailed description of that model is presented in [3].

WEST part

For the WEST part, the system is modeled with the following elements:

- **System dynamics:** represents the dynamics of the WEST part of CE system
- **Split imbalance:** three synchronous generators with 6.3 GW output power in total
- **Defensive trippings:** two loads, with 1.7 GW power in total
- **Load:** represents all the load in the WEST part of CE system

EAST part

For the EAST part, the system is modeled with the following elements:

- **System dynamics:** represents the dynamics of the EAST part of CE system
- **Split imbalance:** three loads with 6.3 GW power in total
- **Defensive trippings:** four synchronous generators, with 5.1 GW power in total
- **Load:** represent all the load in the EAST part of CE system

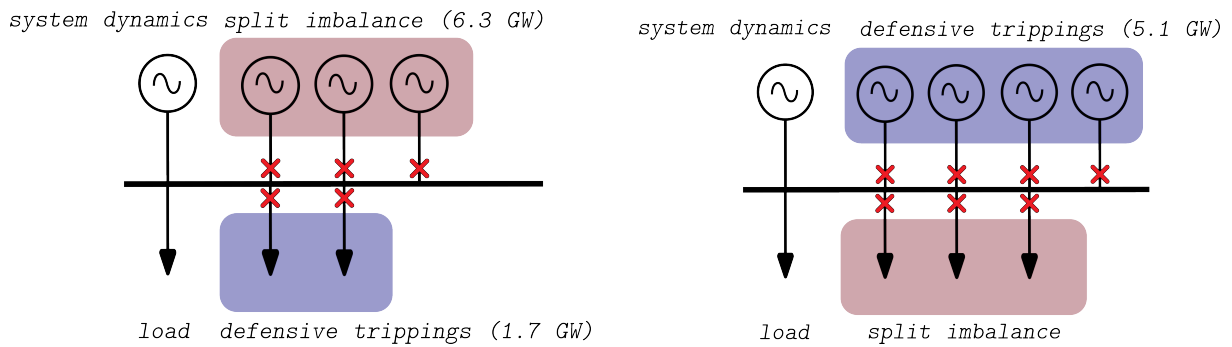


Figure 4 | Initial single node model for the (a) WEST and the (b) EAST part

3.2.2 Model parameters

Proper modeling of a synchronous area requires several data and information from the actual system desired to model. In our case, the most significant parameters from the CE system and the two parts (WEST, EAST) are collected in Table 3. A detailed description about the methods of data collection and calculations are presented in [3].

Table 3 | System simulation parameters [3]

Parameter	CE before split	WEST	EAST
System load (GW)	≈382	≈317	≈65
Rated power (GVA)	≈520	≈434	≈89
Kinetic energy E_k (GWs)	1930	1770	160
Characteristic λ (GW/Hz)	34.6	28.75	5.89

Although the inertia constants are required for the modeling of synchronous generators, they are not explicitly specified in [3]. Therefore, I calculated them, based on (2.10). The calculated values are shown in Table 4.

Table 4 | Inertia constants of WEST and EAST part

	WEST	EAST
Inertia constant [s]	4.08	1.80

These parameters are used in the synchronous generator model presented in Section 4.

3.2.3 Switch events

Split imbalances As shown in Figure 4, the power imbalance caused by the system separation is modeled by disconnecting $P = 6.3$ GW of generation on the WEST side and an equivalent amount of load on the EAST side. These are divided into three distinct synchronous machines and three corresponding loads, respectively. The three generator and load blocks have power values of 4.4 GW, 1.8 GW, and 0.1 GW, respectively. The moments of disconnection for the generators and loads also differ. Without this division in both time and power, the model would not be able to reproduce the original frequency response [3].

Defensive trippings The defensive trippings are simulated by disconnecting $P = 1.7$ GW of load on the WEST side and a $P = 5.1$ GW of generator on the EAST side. These are divided into two distinct loads and four distinct synchronous machines, respectively. The moments of disconnection for defensive trippings also differ.

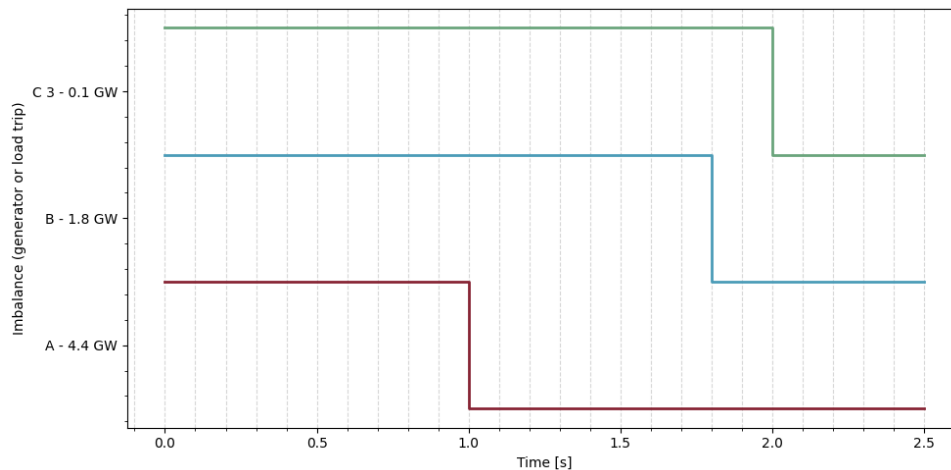


Figure 5 | Disconnection time of machines representing imbalance

3.3 Additional inverter based generations

The initial model accurately reproduces the real frequency response measured on 8 January 2021. However, since then, notable changes have occurred in the generation mix. With the integration of numerous PV and WTG plants, the dynamics and frequency response of the system have evolved significantly, especially during the sunniest hours of summer. To properly simulate the impact of these RES, it is essential to include them in the previously presented single-node model.

The PV and WTG generation is represented by a large-scale PV model and a Type-3 WTG model (Section 6.3), respectively (Figure 6). After connecting these models to the node, the total system load (Table 3) is distributed among the three generators.

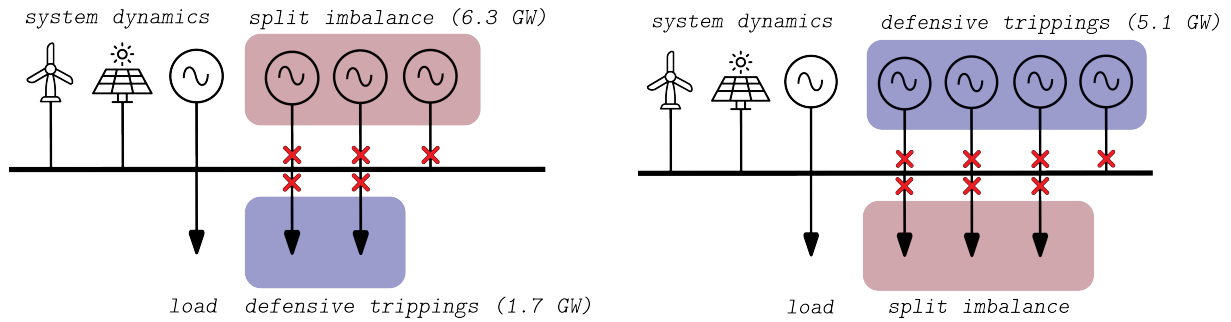


Figure 6 | Upgraded single node model for the (a) WEST and the (b) EAST part

The PV plant and WTG models are introduced in detail in Sections 5 and 6, respectively.

3.4 Simulation methodology

The simulations were performed in DigSilent PowerFactory (PF), which is one of the most widely used software tools to analyse generation, transmission, distribution, and industrial systems.

The chosen method for the simulation was the Root Mean Square (RMS) simulation. There are two main reasons for this:

- The disturbance after system separation lasted 100–150 s, until the frequency stabilized.
- The RMS simulation accurately captures the dynamic response of generators and loads over this timescale, which is essential for reproducing the frequency behavior of the system.

4 Synchronous generator model

4.1 Synchronous generators in general

Synchronous machines are fundamental components of the electric grid, as they traditionally provide most of the power plant feed-in. For this reason, it is essential to model their operation accurately. In the synchronous generators several controllers work together. We have incorporated three such controllers into our model in accordance with ENTSO-E modeling practice [7].

4.2 Control of synchronous generators

4.2.1 Automatic voltage regulator

The Automatic Voltage Regulator (AVR) is an electronic device in synchronous generators that automatically maintains the output voltage at a set value. Its output signal is based on the difference between the reference voltage and the actual voltage measured at the point of connection.

Modeling an AVR can be performed in various ways; in this study a relatively simple model, the Simplified Excitation System (SEXS) excitation model was used [8].

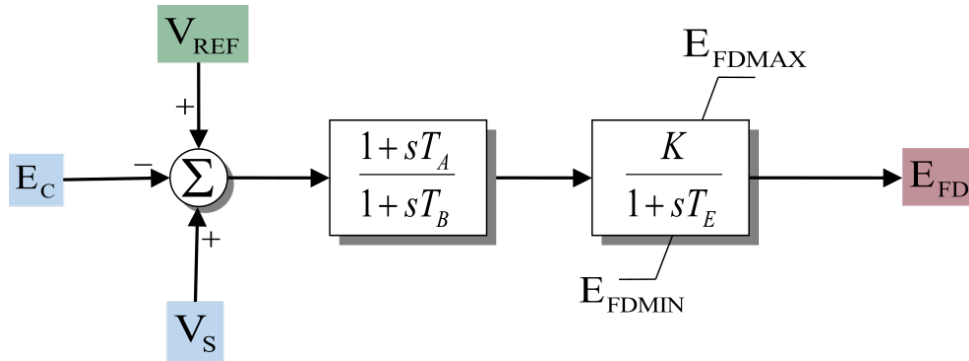


Figure 7 | SEXS controller [8]

4.2.2 Power System Stabiliser

The Power System Stabiliser (PSS) main function is to damp generator rotor oscillations. It actuates through the SEXS AVR; it modulates the output voltage of the synchronous generator. In this study the PSSA2 PSS was used (Figure 8).

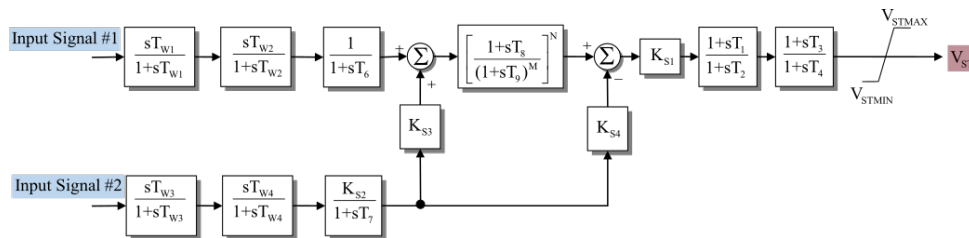


Figure 8 | PSSA2 controller [8]

4.2.3 Turbine governor

The aim of the Turbine Governor (GOV) is to ensure frequency control. If the balance between production and consumption is disrupted, the frequency will change (4.1). Due to the physical properties of the turbine, the rotating mass will provide this missing power, but it will start to rotate more slowly. This is called inertial behavior.

$$P_{gen} - \frac{dE_{kin}}{dt} = P_{load} \quad (4.1)$$

The GOV controller measures the turbine speed and compares it to the frequency. If it detects a difference, it intervenes and increases or decreases the amount of steam/gas entering the turbine based on the feedback signal.

We used the **TGOV1** controller [7] to model the GOV controller.

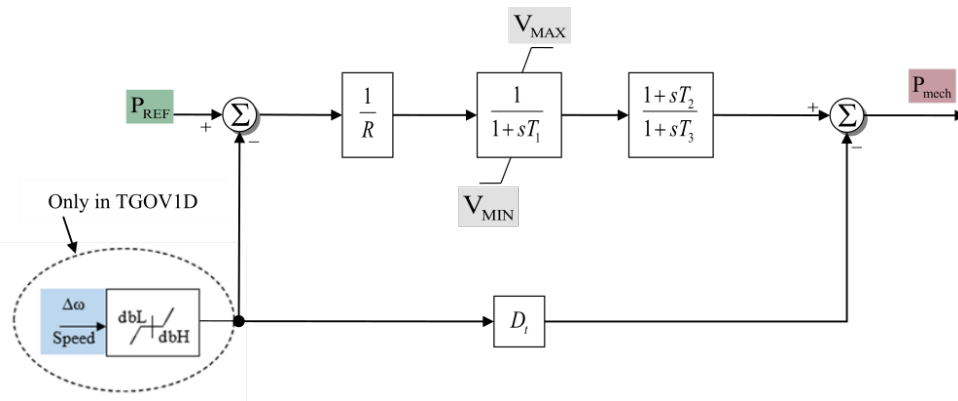


Figure 9 | TGOV1 controler [8]

5 PV plant model

5.1 WECC Large scale PV plant model

In order to properly simulate the PV based production in a one node system, the generic large scale PV plant model developed by Western Electricity Coordinating Council (WECC) was used [9]. This model was previously implemented into PowerFactory (PF) [10] [11]. During our studies this model was used.

The model consists of three main controllers (REPC_A, REEC_B, REGC_A), see Figure 10. Detailed description of these models has been previously reported in [12] [9].

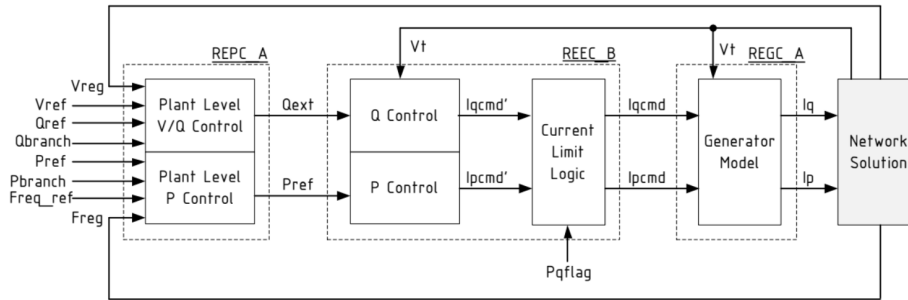


Figure 10 | WECC Large scale PV plant model structure [9]

- **REPC_A:** Renewable Energy Plant Control, responsible for plant level active and reactive power regulation with droop control and Fast Frequency Response (FFR). Detailed presentation is presented in Section 5.1.1.
- **REEC_B:** Renewable Energy Electrical Control, responsible for local active and reactive power control
- **REGC_A:** Renewable Energy Generator/Converter is a high bandwidth current regulator with active and reactive current management

5.1.1 REPC_A control

Since the REPC_A is responsible for plant level active power regulation, its role is emphasized in the case of system split studies, where the proper active control regulation is a cardinal question. The REPC_A controller's active power regulator consists of two main parts: the droop control and the FFR (Figure 11).

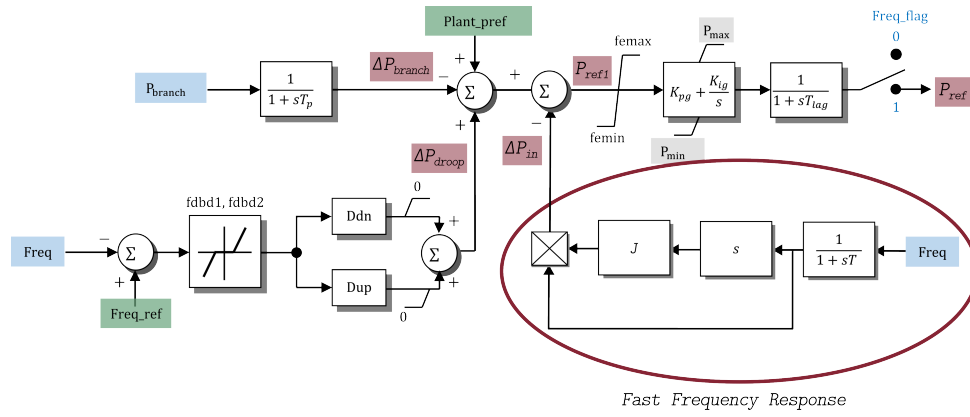


Figure 11 | REPC_A active power control [12]

Droop control

The droop control is an active power regulator logic which output signal ΔP_{droop} is proportional with the **frequency deviation** $\Delta f = Freq - Freq_ref$. Its two parameters are Ddn and Dup , which defines the strength of down and up regulation respectively. The dimension of Ddn and Dup is $[\frac{MW}{Hz}]$.

The droop control is forming part of the original WECC Large scale power plant model[9].

Fast Frequency Response (FFR)

The FFR is an active power regulator logic as well, however its output signal ΔP_{in} is proportional with the **frequency derivative** $\frac{df}{dt}$. Its one parameter J controls the strength of the FFR.

The FFR is not forming part of the original WECC Large scale power plant model [9], it was previously developed as part of our Students' Scientific Conference paper [12].

5.2 De-loaded operation

In the absence of rotational mass, PV plants can't provide synchronous inertial response. Instead of it their active power output is regulated according to the changes in the grid's frequency.

In the case of over-generation, the PV plant simply reduces its active power output. However in the case of generation loss, the solution is not that simple. If the PV plant operates at its maximum value, it won't be able to feed additional extra power towards the grid. Even though it would be crucial.

To address this problem, a simple yet effective solution can be implemented. If a PV plant was operated below its maximum power (de-loaded operation), it would be able to supply additional active power. Which means the PV plant would be able to take part actively in the frequency control processes.

In this study the possible effects of de-loaded PV operation is thoroughly examined [13].

6 Wind turbine generator model

6.1 Reasons of modeling

6.2 Wind turbine types

In order to transform the mechanical power of the wind into electrical power, several methods are implemented [14]. Even so, in the library of Western Electricity Coordinating Council (WECC) generic models, only four of them are included [15]. Thus, in this section, these four main types will be briefly reviewed.

6.2.1 Type1 - Direct-connected SCIG [14]

Direct-connected means that the generator of the wind turbine is connected directly to the grid without any power electronics. In the WECC model library, the Squirrel Cage Induction Generator (SCIG) type generator is modeled.

The main advantages of using SCIGs are low cost, stable performance, small size and robustness. Its main parts are only an induction generator and a turbine, which are connected to each other via a gearbox. In order to maintain the grid frequency, the rotational speed of the turbine and generator is fixed and it is controlled by the pitch angle. The fixed speed wind generator is directly connected to the grid via a soft-starter and a transformer.

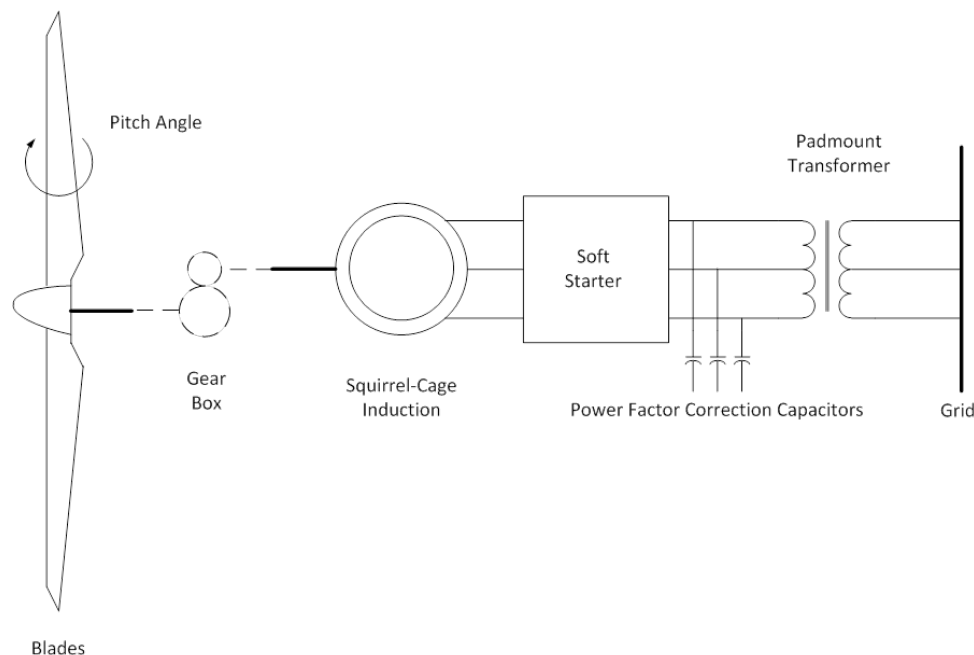


Figure 12 | Direct-connected SCIG [16]

Despite the several advantages of SCIGs, the significant drawbacks are the inability to control reactive power and the lack of fault ride through capabilities, leading to poor stability in the case of voltage dips or disturbances.

6.2.2 Type2 - WRIG with external resistance control [14]

In Wound Rotor Induction Generator (WRIG) design, each rotor winding is electrically accessible via slip rings, allowing for the control of its resistance. In practice, it is performed by power electronics controlled resistance.

The main advantage of the WRIG is that by varying the resistance of the rotor windings, the slip of the generator can be controlled by 2 – 5%. Thus, a small variation in wind turbine speed is acceptable, making it more effective under varying wind conditions.

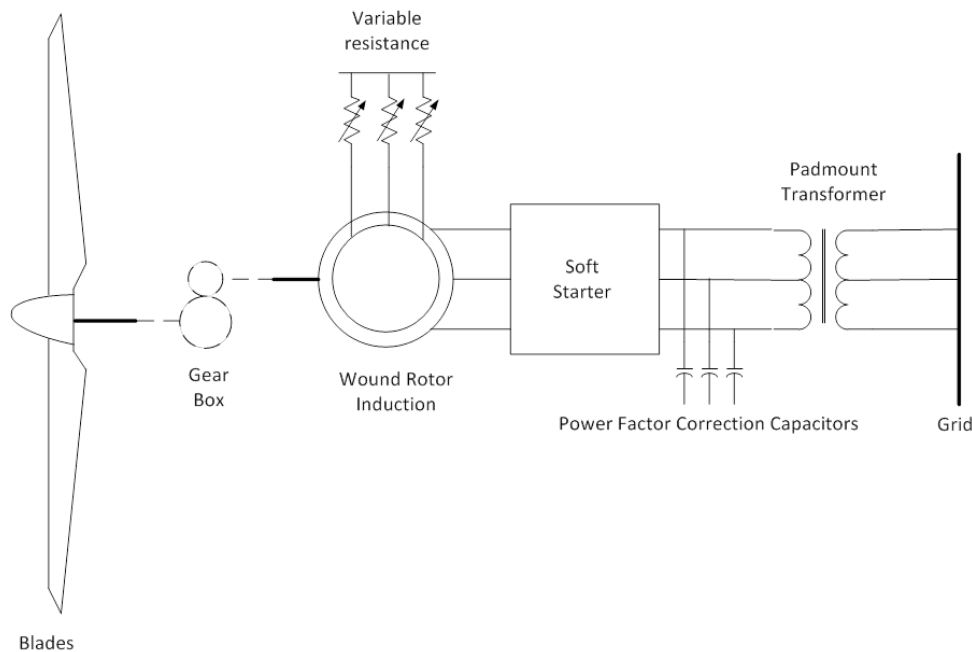


Figure 13 | Direct-connected WRIG [17]

Despite their capability for variable speed operation, WRIGs have a significant reactive power demand and an inability to control reactive power. Furthermore, the additional resistances in the rotor circuits lead to extra losses.

6.2.3 Type3 - Directly-connected DFIG [14]

The Doubly Fed Induction Generator (DFIG) is a wound rotor induction generator whose stator is connected directly to the grid, while its rotor is connected via an AC/AC voltage source type power converter. This arrangement creates the opportunity to directly control the AC signal applied to the rotor windings.

The two main advantages of DFIGs are the following:

- **Variable speed:** with the AC/AC power converter, the frequency and polarity of the rotor signal are adjustable. Thus, the frequency and direction of the rotor's magnetic field are controllable. As a consequence, the frequency of the AC signal generated across the stator could be kept constant under varying rotor speeds.
- **Decoupled control of active and reactive power:** the AC/AC power enables decoupled active and reactive power control

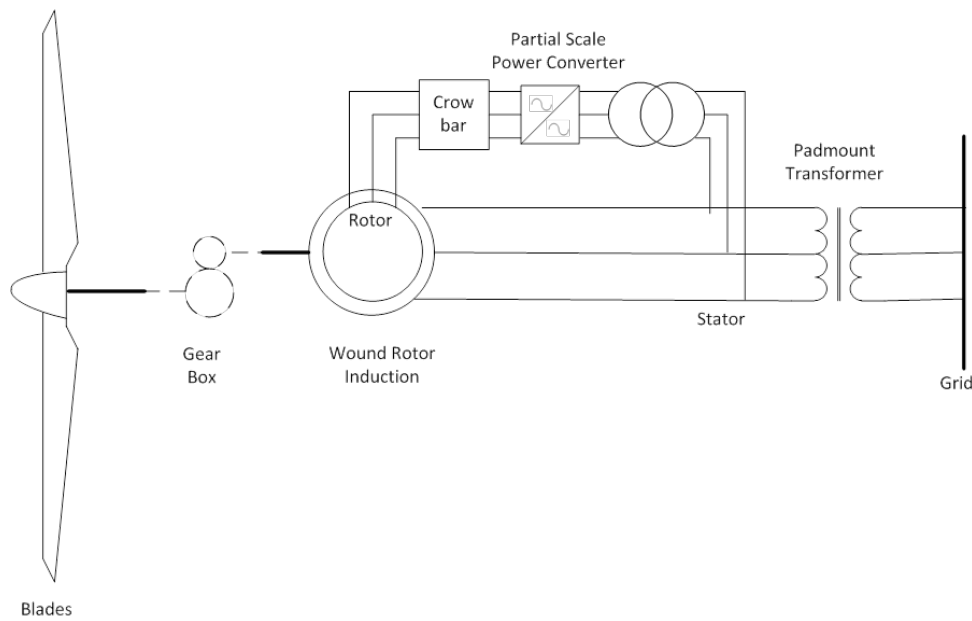


Figure 14 | Direct-connected DFIG [18]

One notable drawback of DFIGs is their poor low-voltage ride-through capabilities.

6.2.4 Type4 - WTG with Full-Rated Power Converter [14]

The generators in this category are connected to the grid through a full-rated AC/AC power converter. The generator could be (a) a WRIG, (b) Wound Rotor Synchronous Generator (WRSG), (c) Permanent Magnet Synchronous Generator (PMSG).

The main advantages of these generators are their outstanding performance in variable wind speed scenarios, high efficiency, reactive power, and voltage control.

The only disadvantage of these types of generators is their poor fault current contribution capability.

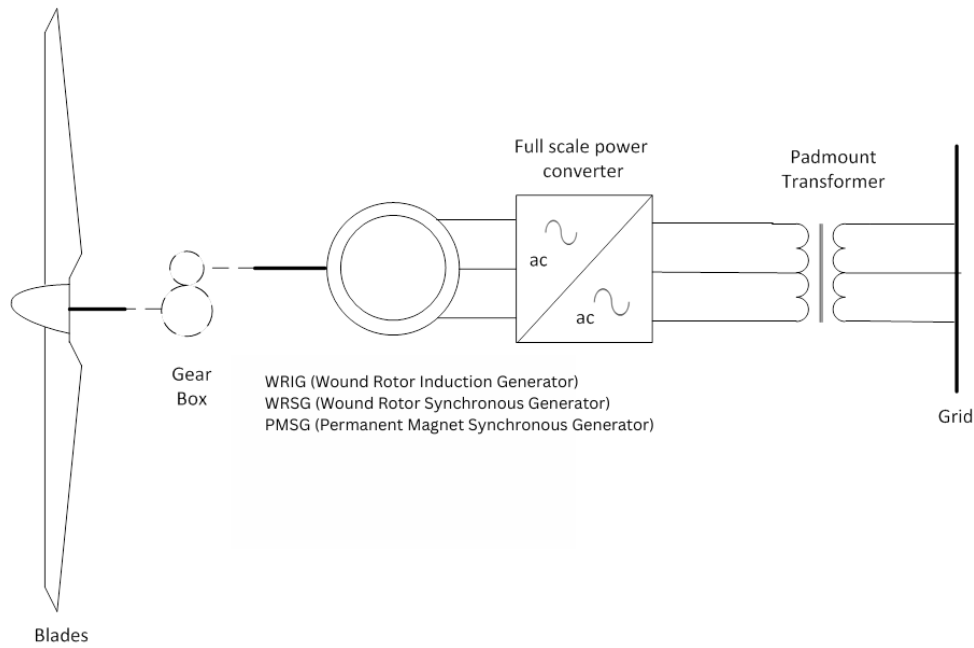


Figure 15 | Direct-connected DFIG [18]

6.3 WECC Type3 WTG model

Although there are no available data on the distribution of the different WTG types, it is assumed that the Type3 wind turbine is among the most widespread ones. Hence our model is based on that type of WTG.

In the literature several WTG models are available, however in PF just two models (*IEC 61400-27-1* and *WECC 2nd Generation Wind Turbine model*) developed by WECC are implemented. A detailed comparison of the two models is reported in [19]. Since the PV systems have been previously modeled by the WECC's model, the choice fell on the WECC 2nd Generation Wind Turbine model [20] [21] [22].

The WECC consists of eight separated control units (Figure 16). The model structure follows the WECC's modular construction method.

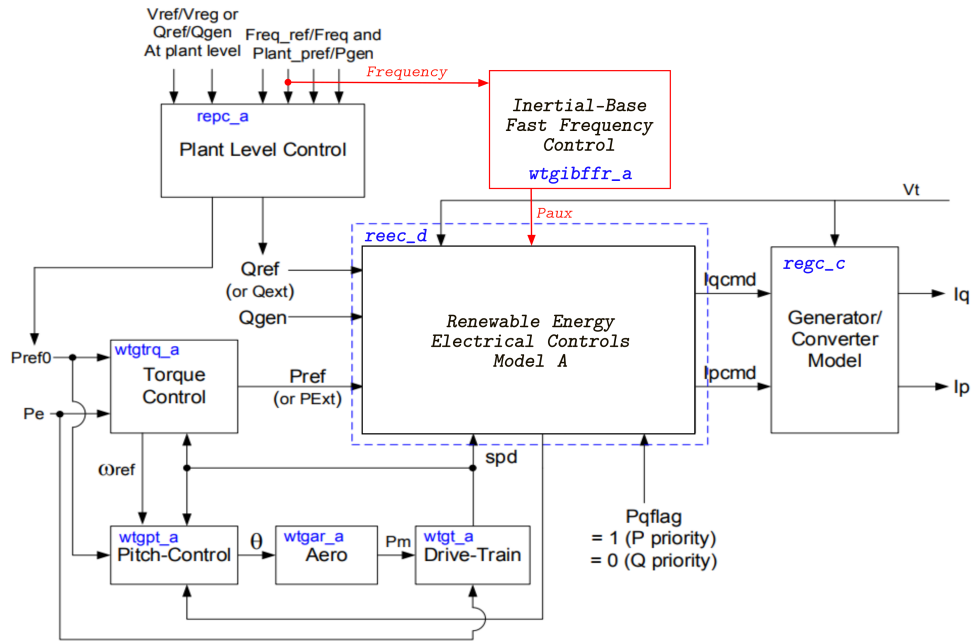


Figure 16 | Control diagram of WECC Type3 WTG model [23]

The eight parts of the model are the following [21]:

1. **REPC_A**: Renewable Energy Plant Control model with plant level active and reactive power control. Its inputs, beside the frequency and active power signals ($Freq_ref$, $Freq$, $Plant_pref$, are reference and measured, either voltage ($Vref$, $Vreg$) or reactive power ($Qref$, $Qreg$) signals. Its outputs are $Qref$ or $Qext$ and $Pref$.
2. **REEC_D**: Renewable Energy Electrical Control model, with inputs: $Pref$, $Qgen$, $Qref$ and feedback of reactive power generated $Qgen$. The outputs are active and reactive current command $Ipcmd$, $Iqcmd$.
3. **REGC_C**: Renewable Energy Generator/Converter model with inputs $Ipcmd$, $Iqcmd$ and outputs of real and reactive current injection Ip , Iq
4. **WTGT_A**: drive-train control unit, it simulates the drive-train oscillation. Its output signal is speed, spd , which is a vector, consists of the angular speed of turbine ω_t and generator ω_g .
5. **WTGAR_A**: simple linear model of the turbine aero-dynamics
6. **WTGPT_A**: pitch controller model, with a parameter K_{cc}
7. **WTGTRQ_A**: simple emulation of torque control

It is worth noting that the first three control units ($REPC_A$, $REEC_D$, $REGC_C$) represent a newer generation of the same type of controllers as those discussed in Section 4.

6.4 Fast Frequency Response in WTGs

In cases of power imbalances, the most significant controlling systems in WTGs are the **droop control** and the Inertial-Base Fast Frequency Response (IBFFR). Both of them control the active power output of the WTG. However, their purposes and methods are distinct. In this subsection, the Fast Frequency Response (FFR) control is presented in detail.

6.4.1 Theory of IBFFR [23] [24]

Despite WTGs being connected to the grid most of the time via power electronic devices, the rotational energy stored within their blades and rotors enables them to contribute to frequency control in an inertial-like manner. To achieve this, in the event of a frequency drop, the rotational energy can be extracted from the rotor and injected into the grid.

Derived from the first law of thermodynamics [25], the total energy of a system should be conserved. In the case of WTGs it means that the total energy produced by the rotor should be equal to the total energy consumed by the electric generator, if the mechanical losses are neglected.

$$\Delta E = E_m - E_e = 0 \Rightarrow E_m = E_e \quad (6.1)$$

Applying the definition of energy ($E = \int P dt$):

$$\int P_m dt = \int P_e dt \quad (6.2)$$

And express it with torque:

$$\int T_m \cdot \omega_m dt = \int T_e \cdot \omega_e dt \quad (6.3)$$

The equations (6.2) and (6.3) are valid until the grid is perfectly balanced. At the moment of any disturbance, the momentary values of mechanical and electrical power and torque are not in equilibrium. The difference is absorbed or released by the rotor's stored kinetic energy (6.4).

$$\int T_m \cdot \omega_m dt + \Delta E_{rot} = \int T_e \cdot \omega_e dt \quad (6.4)$$

Where ΔE_{rot} is the change in the rotational energy of the rotor.

$$E_{rot} = \frac{1}{2} J \omega_m^2 \quad (6.5)$$

Combining equations (6.4) and (6.5), the following equation can be expressed:

$$\int T_m \cdot \omega_m dt + \frac{1}{2} J \cdot \Delta \omega_m^2 = \int T_e \cdot \omega_e dt \quad (6.6)$$

Assuming that after a disturbance, the torques (both electrical and mechanical) are constant over time, the only variable that can balance the energy deviation is the angular speed (ω). Therefore, in the case of *generation loss*, the values of electric power and torque *exceed* those of mechanical power and torque. Hence, in order to satisfy the law of conservation of energy, the mechanical angular speed (ω_m) shall decrease, extracting additional power from the rotor.

6.4.2 Constraints of IBFFR

Even though IBFFR is a practical method for stabilizing frequency in the case of grid disturbances, it has three major limitations:

- **Extractable energy:** theoretically the maximum extractable energy from the rotor is the actual kinetic energy of the rotor. However in reality, in favor of secure operation, stalling of the rotor is not permissible.
- **Extractable power:** The extractable power from the rotor is typically limited to about 5 to 10% of the actual power output of the WTG [23].
- **Activated duration:** the energy and power constraints define a maximum activated duration for the IBFFR control.

6.4.3 Recovery

The recovery is a special property of IBFFR control in WTGs. It basically means that after the decrease of the rotor's angular speed, it should accelerate back to its original value. This acceleration requires time and a decrease in electrical power output. During this recovery period, the WTG's contribution to frequency stability is even less than it was before the IBFFR intervention, which could have severe effect on grid stability.

The key of the recovery is that the recovered energy should be equal or slightly greater (because of mechanical losses) than the extracted energy from the rotor (Figure 17. Area A and B).

6.5 Fast Frequency Response in Type3 WTG model

In order to properly model the IBFFR, the EPRI coordinating with WECC has developed a generic IBFFR control loop, called WTGIBFFR_A (Wind Turbine Generator Inertial Based Fast Frequency Response) [23].

6.5.1 The basics of WTGIBFFR_A

The WTGIBFFR_A control is based on the characteristic curve shown on Figure 17.

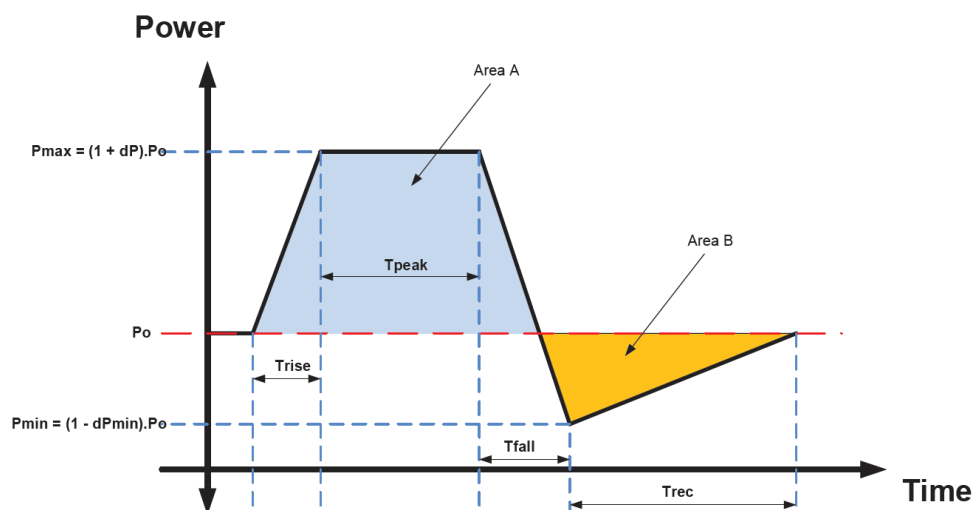


Figure 17 | Characteristic of IBFFR [23]

The explanation of variables mentioned on Figure 17. are the following:

- P_o : initial output power of WTG
- P_{max} : maximum output power of WTG during IBFFR (Section 6.4.2.)
- P_{min} : minimum output power of WTG during recovery period
- dP : additional power during FFR
- dP_{min} : maximum of power deviation during the recovery peiod
- T_{rise} : time to increase output power of WTG from P_o to P_{max}
- T_{peak} : sustained time of P_{max} (Section 6.4.2.)
- T_{fall} : time to decrease output power of WTG from P_{max} to P_{min}
- T_{rec} : time of recovery (Section 6.4.3.), calculated according to (6.7)

$$T_{rec} = \frac{2 dP}{dP_{min}} \left[\frac{T_{rise}}{2} + T_{peak} + \left(\frac{dP}{dP + dP_{min}} \right) \frac{T_{fall}}{2} \right] - T_{fall} \left[1 - \frac{dP}{dP + dP_{min}} \right] [s] \quad (6.7)$$

For accurate dynamic simulation, the WTGIBFFR_A model defines six individual values for each of the parameters described above. This level of detail ensures that the WTG behavior is represented precisely across the entire operating range of P_o , with each interval captured by its corresponding parameter set.

6.5.2 The control loop of WTGIBFFR_A

The above presented (Section 6.5.1.) logic is implemented in the WTGIBFFR_A control loop as it shown on Figure 18.

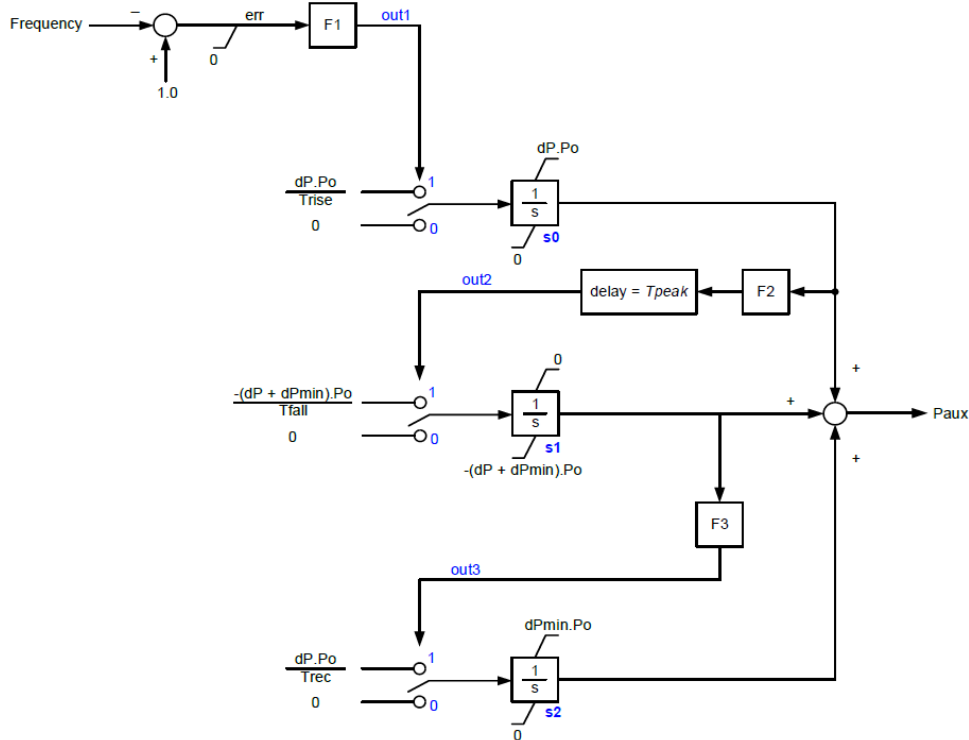


Figure 18 | Control loop of IBFFR [23]

The explanation of each part of Figure 18. are the following:

- **F1:** represents a simple logic which activates the FFR if the frequency deviation is large enough (6.8)
- **F2:** this function picks the right parameter values based on P_o

$$\text{if } err \leq dbd \text{ then } out1 = 0, \text{ else } out1 = 1 \quad (6.8)$$

Although it is sometimes called inertia based FFR (IBFFR), there is no inertia like response during the FFR process. In the case of WTGs FFR is purely a matter of control, while in traditional synchronous machines it is an inherent property.

7 Case studies and discussion

The case studies presented below were simulated exclusively on the *WEST* single-node model described in Section 3. In total, three different scenarios were examined based on the types of generators in operation:

1. Synchronous generator: the case presented in [3], gives the values measured during the real life system split
2. Synchronous generator + PV
3. Synchronous generator + WTG
4. Synchronous generator + PV + WTG

In each scenario, several different sub-scenarios were simulated by varying some of the model parameters.

7.1 Synchronous generator

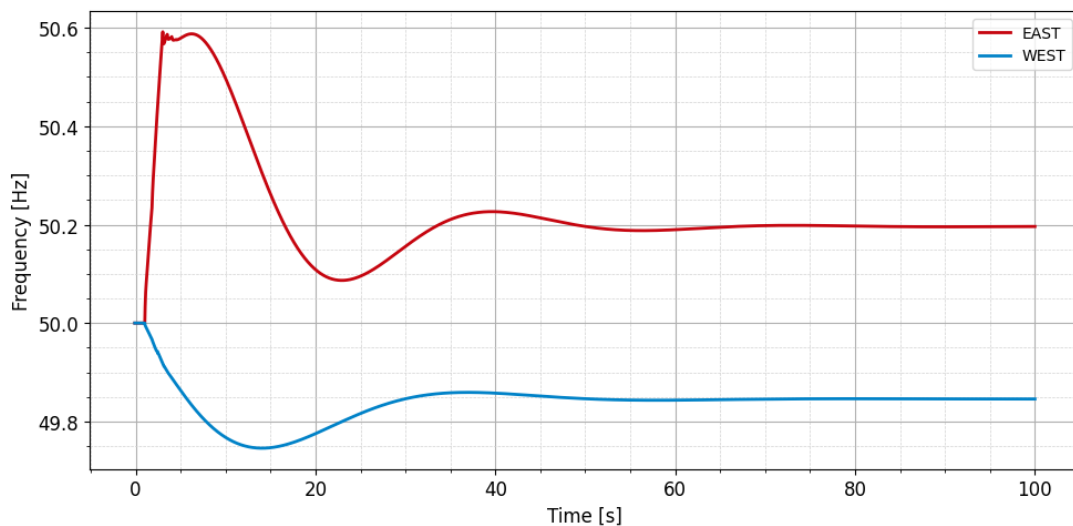


Figure 19 | Frequency response of WEST and EAST part in case of 100% synchronous generation

In the following scenarios, these frequency responses serve as a point of comparison. These values are shown in each frequency response plot.

7.2 Synchronous generator and PV

During these simulations, only the synchronous generator and the PV plant were active from the three generators shown in Figure 6.

The simulations were based on the changes in the following parameters:

- PV penetration ratio [%]

- Strength of FFR (J [p.u.]): As mentioned in (2.5), J physically represents the rotational inertia of a rotating machine. However, in the FFR control of the WECC large-scale PV plant, it is implemented in per-unit form, where the base value is the nominal system frequency.

By varying these variables, 16 + 1 cases (17 in total) were simulated (Figure 20). The +1 refers to the case when there was no PV generation on the grid,

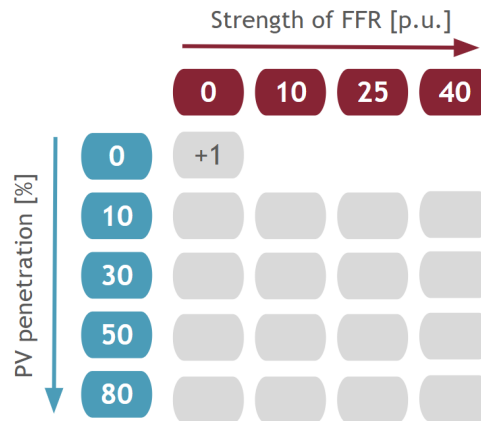


Figure 20 | Case studies

7.2.1 10 % PV penetration

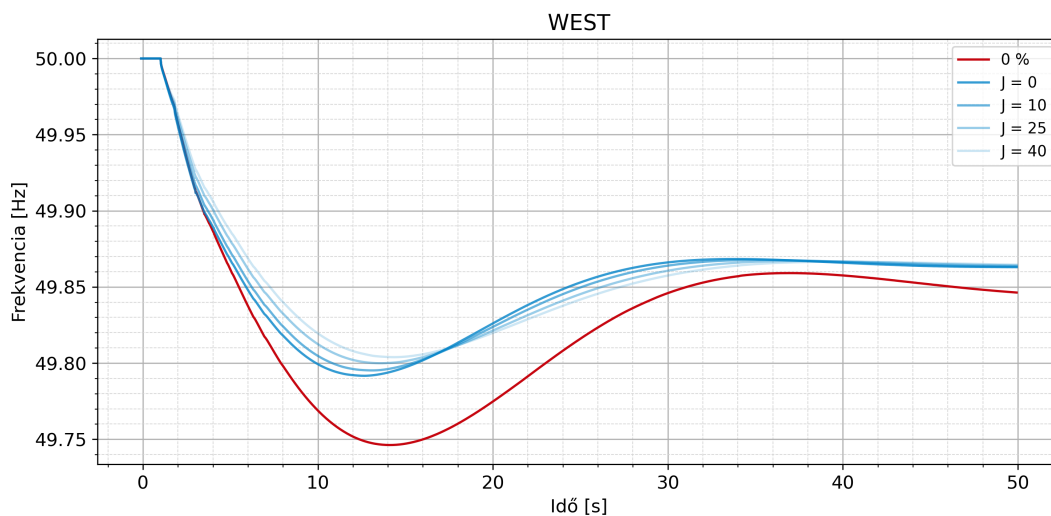


Figure 21 | Frequency response WEST - 10% PV penetration

According to Figure 21 the following conclusions can be deduced: the frequency nadir increased by approximately $\sim 0.05 - 0.07$ Hz, compared to the case where there was no PV generation in the grid.

7.2.2 30 % PV penetration

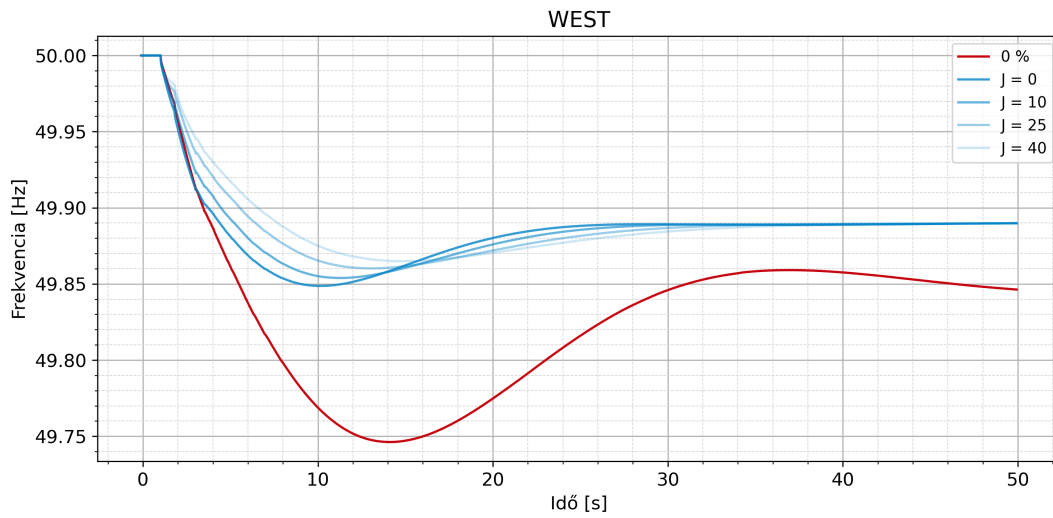


Figure 22 | Frequency response WEST - 30% PV penetration

As shown in Figure 22, the maximum frequency deviation after the disturbance has decreased significantly compared to the results in Figure 21 and the original frequency response (red line). The frequency nadir increased around $\sim 0.1 - 0.15$ Hz compared to the original frequency response.

7.2.3 50 % PV penetration

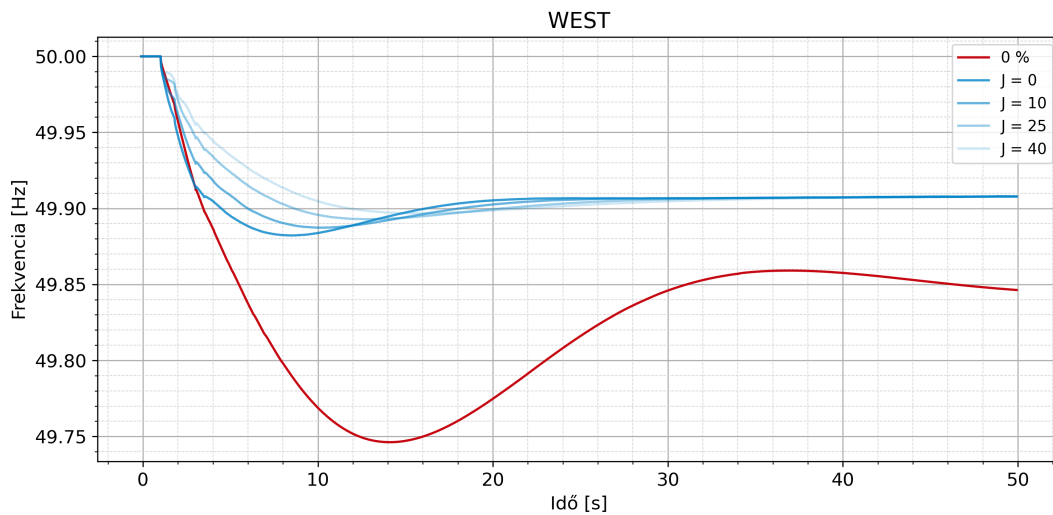


Figure 23 | Frequency response WEST - 50% PV penetration

As shown in Figure 23, the frequency deviation further decreased. The frequency nadir increased by $\sim 0.13 - 0.16$ Hz compared to the original frequency response.

7.2.4 80 % PV penetration

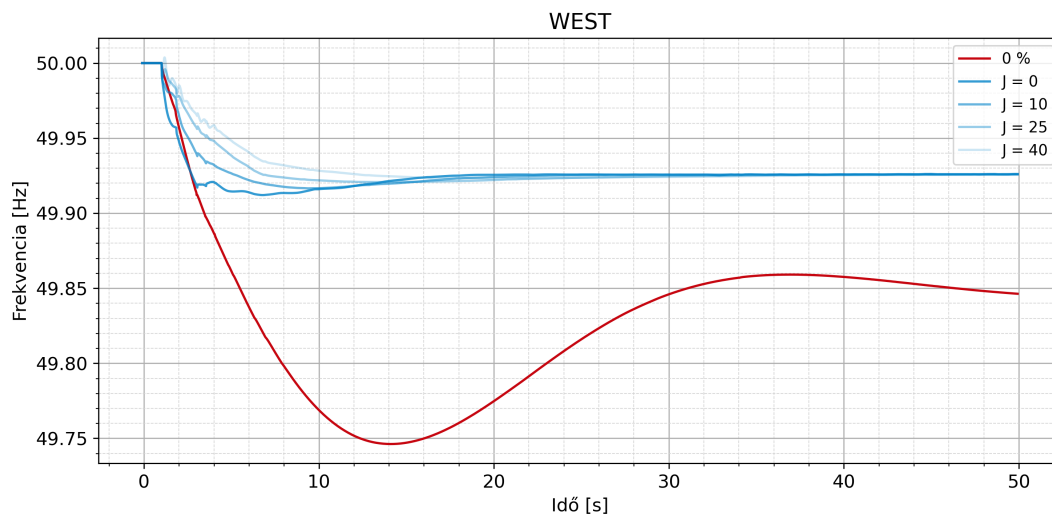


Figure 24 | Frequency response WEST - 80% PV penetration

In Figure 24, it can be seen that the frequency nadir increased even further; however, the frequency response began to oscillate, especially for larger J values.

7.2.5 General effect of PV generation

From Figures 21, 22, 23, and 24, it is clear that increasing the strength of the FFR (J) in the modelled PV plant, improves the system's frequency response, at each penetration level.

To sum up, it can be concluded that increasing the PV penetration level and the strength of FFR improves the frequency response of the system. This tendency is shown in Figure 25, where the maximum frequency deviation is plotted as a function of the PV penetration level and the FFR strength.

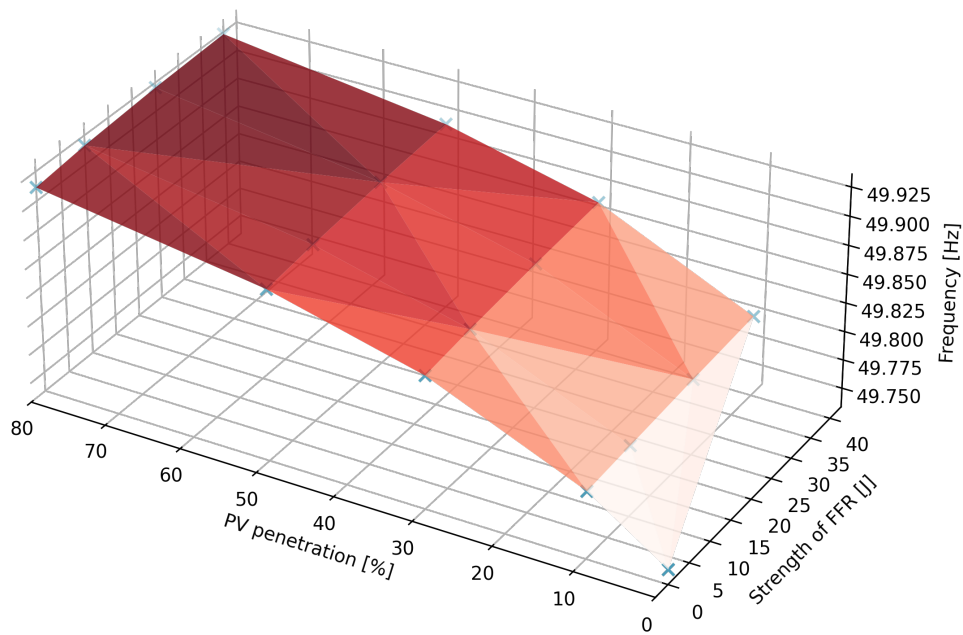


Figure 25 | Maximum frequency deviation as a function of PV penetration and strength of FFR

7.3 Synchronous generator and WTG

In this arrangement, only the value of WTG penetration was changed, using the same four steps as for the PV case (Figure 20). During these studies, none of the WTG parameters were modified; furthermore, the parameters of the most important control unit in terms of disturbance response the WTGIBFFR_A control were configured according to [26].

The results of the simulations are shown in Figure 26.

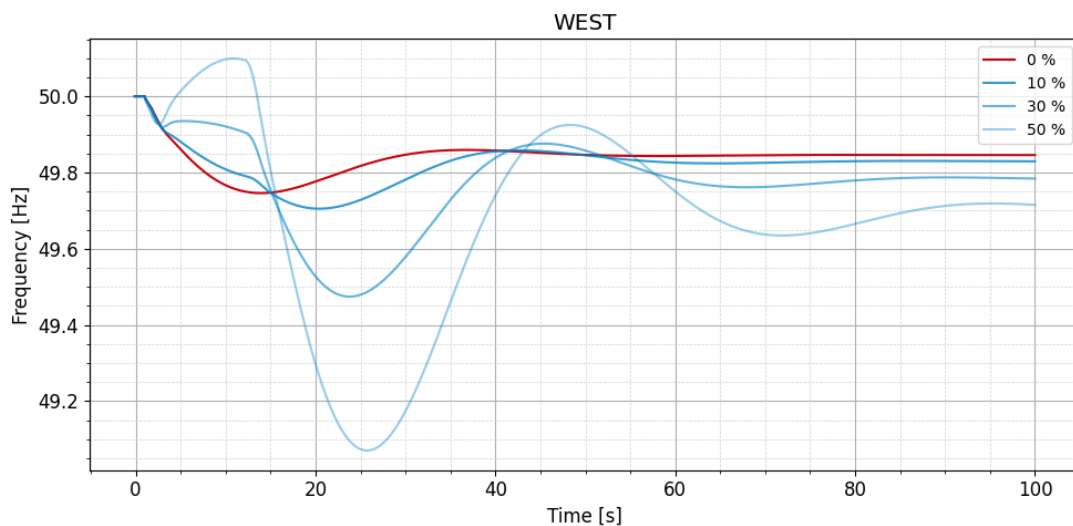


Figure 26 | Frequency response in case of different WTG penetration

7.3.1 The effect of wind turbines

In the following, the four key observations regarding Figure 26 are presented:

- **Sudden fracture in the frequency plot:** Approximately three seconds after the disturbance, an abrupt change appears in the frequency trajectory. At 10% WTG penetration, this fracture results in a further decrease in the slope of the frequency response, while at 30% and 50% penetration levels, the frequency begins to rise; in the latter case, it even exceeds the nominal value.
- **Sudden drop of frequency:** Although the frequency appears to stabilise between three and twelve seconds, it subsequently experiences a pronounced decrease.
- **Extreme frequency deviations:** The maximum frequency deviation exceeds the value observed in the original configuration (with only synchronous machines). The magnitude of the deviation increases as the ratio of wind turbines increases. At 80% penetration, the maximum frequency deviation reaches approximately 8.7 Hz.
- **Longer oscillations:** Increasing the penetration level of WTG leads to larger and more sustained oscillations in the system frequency.

7.3.2 Main causes behind the altered frequency response

The possible reasons behind the earlier mentioned behavior, are the followings:

- **Sudden fracture in the frequency plot:** The abrupt change is caused by the FFR implemented in the WTG control. When the frequency exceeds a predefined threshold (in our model $f_{\text{nominal}} + f_{\text{nominal}} \cdot 0.05$), the FFR activates, and after a specified rise time (T_{rise}), the output power increases to a predefined level (P_{max}). This process occurs within a few seconds, resulting in a sudden change in output power and system frequency.
- **Sudden drop of frequency:** The WTG can output P_{max} only for a limited duration (T_{peak}). In practice, this duration depends on the physical characteristics of the turbine (Section 6), while in the simulation it is assigned as a model parameter. When this period ends, the output power drops to P_{min} , initiating the recovery process and causing a significant frequency deviation.
- **Extreme frequency deviations:** These occur as a direct consequence of the recovery period.
- **Longer oscillations:** The large frequency oscillations originate from the dynamics of the synchronous generators, likely triggered by the sudden and substantial frequency disturbances in the system.

To sum up the effect of WTGs on the frequency response, the following statement can be concluded:

The presence of Wind Turbine Generators equipped with properly tuned Fast Frequency Response can improve the frequency response during the first 10–12 seconds following the disturbance. However, beyond this period, their contribution to frequency stability becomes negative, especially at higher penetration levels.

It is worth mentioning that the frequency response shown in Figure 26 is not representative of a real network configuration. The magnitude and steepness of the frequency deviations are not permissible in actual electric grid operation; therefore, further development of the model is needed. Nevertheless, the results illustrate the basic characteristics of FFR in WTGs.

7.4 Synchronous generator, PV and WTG

This arrangement is intended to simulate the typical behaviour of synchronous machines, PV plants, and WTGs in the event of system separation. Unlike the previous scenarios, in this case the penetration levels of WTGs and PV plants are based on measured data provided by the ENTSO-E Transparency Platform [27], Table 5. The data were collected between 1 June 2025 and 31 August 2025 from the entire European synchronous zone (Figure 1).

For the study, three different scenarios were selected:

1. **Maximum PV generation:** the period when PV generation reached its maximum.
2. **Maximum combined PV and wind generation:** the period when the sum of PV and wind generation was highest.
3. **Maximum wind generation:** the period when wind generation reached its maximum.

Although the data were collected from real measurements, the simulation does not aim to represent a realistic system disturbance.

Table 5 | Generation values by type

Scenario	PV ratio	Wind ratio
Maximum PV	51.6%	4.2%
Maximum PV + wind	51.6%	4.2%
Maximum wind	15.7%	25.4%

The simulations were performed using two different approaches. The difference between the two methods lies in the operating mode of the PV system. In the first approach, the PV plant was operated in normal mode (Figure 27), whereas in the second approach, it was operated in de-loaded mode (Figure 28).

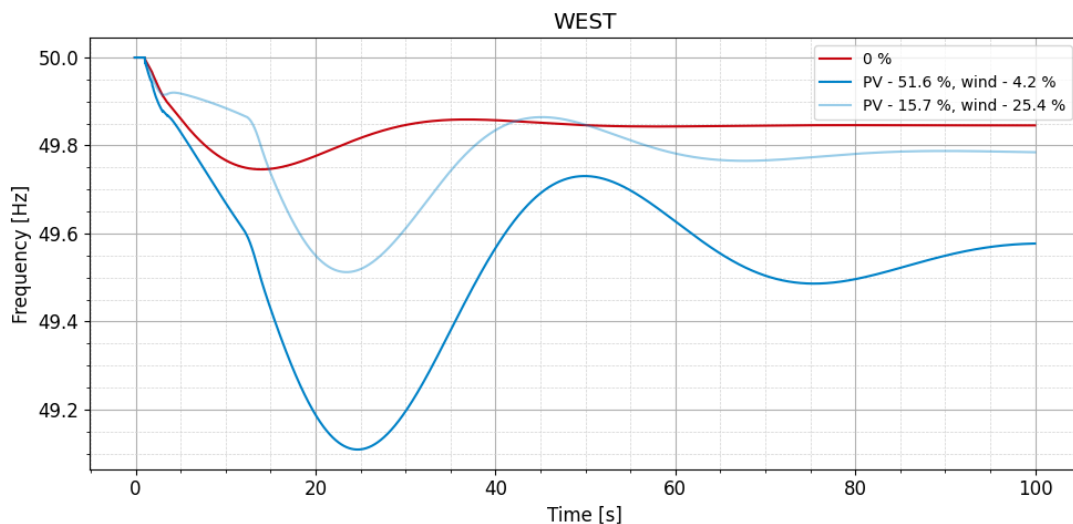


Figure 27 | Frequency response with synchronous generator, WTG and PV with no de-loaded operation

In Figure 27, at a high PV penetration level and without de-loaded operation, the maximum frequency deviation increases to $\Delta f = 0.9$ Hz. Such a deviation is not acceptable in electric grid operation. On the other hand, when the PV penetration is relatively lower, the frequency

deviation decreases, and due to the high WTG penetration, the effect of FFR can be observed between 3 and 12 seconds.

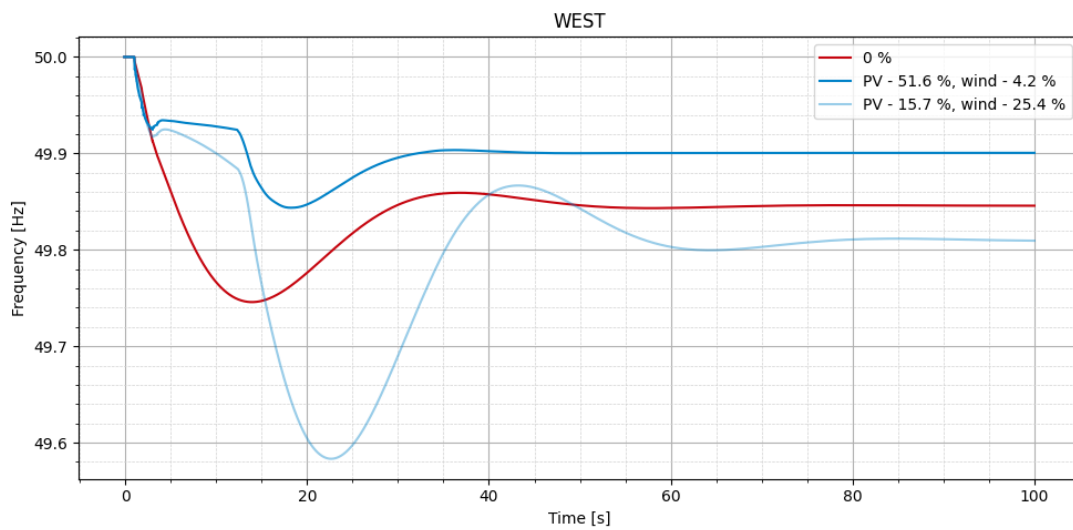


Figure 28 | Frequency response with synchronous generator, WTG and PV with de-loaded operation

After switching back to de-loaded operation, the frequency response of the high PV penetration scenario improved significantly, with the maximum frequency deviation decreasing to 0.16 Hz. Furthermore, a slight improvement can also be observed in the low PV penetration case. The effect of the WTG FFR is clearly visible and highly pronounced in all scenarios.

7.5 Effect of de-loaded PV operation

As mentioned earlier (Section 5.2), during the case studies the PV plant was operated in a de-loaded mode. Thus, it was capable of effectively stabilizing the frequency. However, if the PV plant in the model operates in normal mode at its nominal power, it cannot provide additional power output in the event of a system disturbance. To illustrate this, a simulation was performed with the PV plant operating in normal mode. The results are shown in Figure 29. Each line represents a different PV penetration level from 0% to 80%.

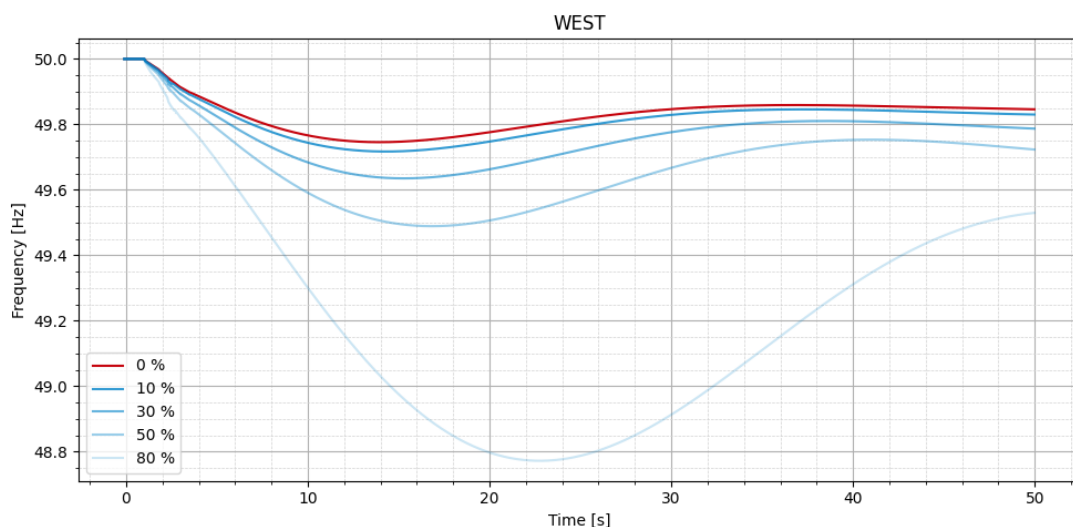


Figure 29 | Frequency response with no de-loaded operation

Comparing Figure 29 with the figures shown in Section 7.2, the differences in the frequency responses are clear. Thus, considering de-loaded operation as a potential operating method to support frequency stability—especially when generation is predominantly inverter-based—would be advisable.

8 Conclusion

8.1 Aims of the research

The aim of the current thesis was to develop and test a single-node model, which consists of the three most common types of power generation units: a synchronous generator, a photovoltaic plant, and a Wind Turbine Generator. The model was successfully built, and several study cases were examined to reveal the fundamental behavior patterns of each generation unit.

8.2 Methodology

Each generation unit was modeled using widely accepted and industrially recognized models. For the synchronous machine, an AVR, a PSS, and a GOV were used. The PV plant was modeled using the WECC Large-Scale Generic PV Plant model. The Type 3 WTG model, developed by WECC, was used to represent the Wind Turbine Generator.

The test scenarios were carefully selected to capture all aspects of the given generation unit's behavior. First, the two renewable energy sources were examined separately, one by one; subsequently, their joint behavior was tested.

In addition, a new approach to the operation of PV plants, namely the de-loaded operation, was examined.

8.3 Conclusion

Based on the performed simulations, the following conclusions can be drawn:

- The operational impacts of PV plants and WTGs are complex and depend on several factors. However, with proper control, they are capable of contributing to the frequency stability of the grid, and in some cases, even outperform the frequency response of traditional synchronous generator-based systems.
- With de-loaded operation, PV plants are capable of contributing to the Frequency Control Reserve (FCR), exhibiting a faster response than traditional synchronous generators. Therefore, further research on their industrial application is recommended.

In summary, I hereby declare that the initial aims of the thesis were successfully accomplished, and the developed model — with slight supplementary improvements to the WTG model — is ready for use in the next phase of the research project.

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